

National University of Technology



Computer Science Department

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Program: Artificial intelligence

Course: Programming for AI Lab

Course Code: CS283

Lab Report- 08

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Julia Programming

1. Hello, World!

```
println(" Hello , Julia !")
```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t1.jl
>>
Hello , Julia !
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> █
```

2. Variables & Constants

```
x = 5

const Y = 10

x += 1

println("x = $x, Y = $Y")
```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t2.jl
>>
x = 6, Y = 10
```

3. Arithmetic Operations

```
a = 10

b = 3

println("Sum: $(a + b)")

println("Product: $(a * b)")
```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t3.jl
>>
Sum: 13
Product: 30
```

4. Conditionals (if-else)

```
n = -5

if n > 0
    println("Positive")
elseif n < 0
    println("Negative")
else
    println("Zero")
end
```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t4.jl
>>
Negative
```

5. Loops (for, while)

```
for i in 1:3

    println("Iteration $i")

end

counter = 1
```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t5a.jl
>>
Iteration 1
Iteration 2
Iteration 3
```

```
counter = 1

while counter <= 3

    println("Counter: $counter")

    global counter += 1

end
```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t5b.jl
>>
Counter: 1
Counter: 2
Counter: 3
```

6. Functions & Control Flow

```
function square(x)

    return x^2

end

println("Square of 5: $(square(5))")
```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t6.jl
>>
Square of 5: 25
```

7. Multiple Dispatch

```
add(x::Int, y::Int) = x + y

add(x::String, y::String) = string(x, y)

println(add(3, 5))
```

```
println(add("Hello ", "Julia"))
```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t7.jl
>>
8
Hello Julia
```

8. Arrays

```
arr = [10, 20, 30]

println("First element: $(arr[1])")

push!(arr, 40)

println("Updated array: $arr")
```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t8.jl
>>
First element: 10
Updated array: [10, 20, 30, 40]
```

9. Tuples & Dictionaries

```
tup = (1, "apple", 3.14)

println("Tuple: $tup")

dict = Dict{"name" => "Alice", "age" => 30}

println("Age: $(dict["age"])")
```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t9.jl
>>
Tuple: (1, "apple", 3.14)
Age: 30
```

10. Structs

```
struct Point
```

```

    x::Float64

    y::Float64

end

p = Point(1.0, 2.0)

println("Coordinates: ($(p.x), $(p.y))")

```

```

PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t10.jl
>>
Coordinates: (1.0, 2.0)

```

11. try-catch

```

try

    println(10 / 0)

catch e

    println("Error: $e")

end

```

```

PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t11.jl
>>
Inf

```

12. Read/Write Files

```

open("data.txt", "w") do file

    write(file, "Julia is fast!")

end

content = read("data.txt", String)

println("File content: $content")

```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t12.jl
>>
File content: Julia is fast!
```

13. DataFrames.jl

```
using DataFrames
```

```
df = DataFrame(Name=["Alice", "Bob"], Age=[25, 30])
```

```
println(df)
```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t13.jl
>>
2x2 DataFrame
  Row | Name      Age
      | String  Int64
-----|-----
  1  | Alice      25
  2  | Bob        30
```

14. Plots.jl

```
using Plots
```

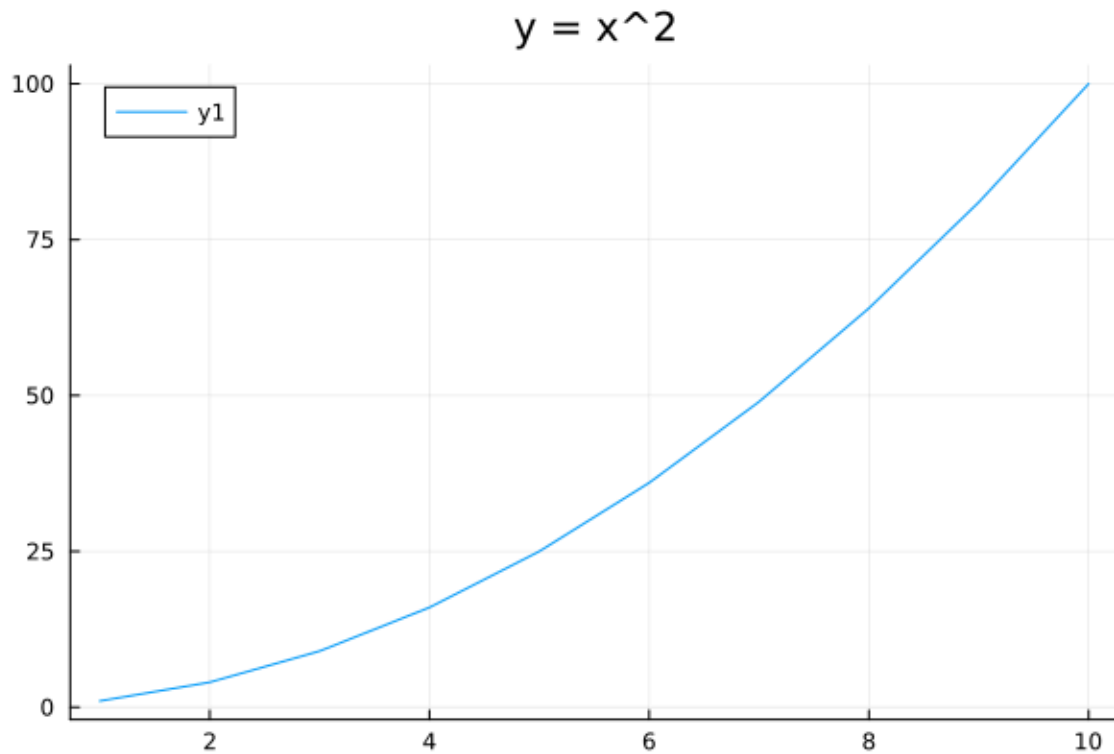
```
x = 1:10
```

```
y = x .^ 2
```

```
plot(x, y, title="y = x^2")
```

```
savefig("plot.png")
```

```
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia tben.jl
>>
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> 
```



15. Flux.jl

```
using Flux

using Random

# Set random seed for reproducibility
Random.seed!(123)

# Create a simple linear model
model = Dense(1 => 1)

# Manually set initial weight to ensure positive slope
model.weight .= Float32[1.0]

model.bias .= Float32[0.0]
```



```

println("Initial model parameters: Weight = $(model.weight), Bias =
$(model.bias)")

# Create training data with consistent Float32 type

X = Float32[1.0, 2.0, 3.0]
Y = Float32[2.0, 4.0, 6.0]

# Reshape inputs to column vectors with correct dimensions
data = [(reshape([X[i]], 1, 1), reshape([Y[i]], 1, 1)) for i in 1:3]

# Define loss function
function loss(m, x, y)

    return Flux.mse(m(x), y)
end

# Calculate total loss over the dataset
function total_loss()

    sum(loss(model, x, y) for (x, y) in data)
end

# Set up optimizer with lower learning rate for stability
opt = Flux.setup(ADAM(0.01), model)

# Training loop with more epochs
num_epochs = 1000

println("Starting training...")

```

```

for epoch in 1:num_epochs

    # Calculate gradients and update weights

    for (x, y) in data

        grad = gradient(m -> loss(m, x, y), model)

        Flux.update!(opt, model, grad[1])

    end

    # Print progress every 100 epochs

    if epoch % 100 == 0 || epoch == 1

        current_loss = total_loss()

        println("Epoch $epoch: Loss = $current_loss, Weight =
$(model.weight[1]), Bias = $(model.bias[1])")

    end

end

println("\nTraining complete!")

println("Final model parameters: Weight = $(model.weight[1]), Bias =
$(model.bias[1])")

# Test the model on training data

println("\nResults on training data:")

for (x, y) in data

    predicted = model(x)

    println("Input: $(x[1]), Target: $(y[1]), Prediction:
$(predicted[1])")

```

```

end

# Test with new data

test_input = reshape(Float32[5.0], 1, 1)

prediction = model(test_input)

println("\nTest with new input:")

println("Input: 5.0, Expected: 10.0, Prediction: $(prediction[1])")

PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> julia t15.jl
>>
Initial model parameters: Weight = Float32[1.0;;], Bias = Float32[0.0]
Starting training...
Epoch 1: Loss = 12.900629, Weight = 1.0275356, Bias = 0.029269237
Epoch 100: Loss = 0.15048791, Weight = 1.734355, Bias = 0.5871269
Epoch 200: Loss = 0.05278124, Weight = 1.8428013, Bias = 0.34785601
Epoch 300: Loss = 0.012392631, Weight = 1.9238623, Bias = 0.16859208
Epoch 400: Loss = 0.0019113566, Weight = 1.9701079, Bias = 0.06622056
Epoch 500: Loss = 0.0001852657, Weight = 1.9906963, Bias = 0.020619765
Epoch 600: Loss = 1.0634409e-5, Weight = 1.9977717, Bias = 0.0049410374
Epoch 700: Loss = 3.350977e-7, Weight = 1.9996047, Bias = 0.00087704224
Epoch 800: Loss = 5.3453846e-9, Weight = 1.99995, Bias = 0.00011065136
Epoch 900: Loss = 4.2120973e-11, Weight = 1.9999956, Bias = 9.850616e-6
Epoch 1000: Loss = 4.1495696e-12, Weight = 1.9999983, Bias = 3.2158985e-6

Training complete!
Final model parameters: Weight = 1.9999983, Bias = 3.2158985e-6

Results on training data:
Input: 1.0, Target: 2.0, Prediction: 2.0000014
Input: 2.0, Target: 4.0, Prediction: 3.9999998
Input: 3.0, Target: 6.0, Prediction: 5.9999986

Test with new input:
Input: 5.0, Expected: 10.0, Prediction: 9.999994
PS C:\Users\Student\Desktop\PFAI LAB 8 JULIA> 

```